TENSORIZED KNOWLEDGE UNITS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

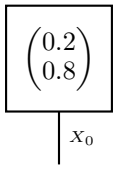
October 12, 2026

ABSTRACT

We investigate, whether tensor networks could serve as a knowledge representation format for the project Alexandria Schuhmann et al. (2025). To represent uncertainties and relations between claims we apply the formalism discussed in Goessmann et al. (2026).

1 Incorporating uncertainties

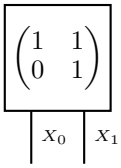
Climate change is happening with 80% *probability.*



2 Modelling relations between claims

When the polar ice is melting, then climate change is happening.

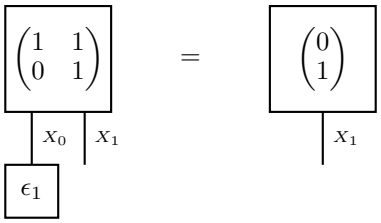
- X_0: Polar ice is melting
- X_1: Climate change is happening



3 Inference

Assuming that the polar ice is melting, what is the probability of climate change, according to the paper?

This would be a contraction with $\epsilon_1 [X_0]$



4 Similarity by contractions

The extracted tensorized Knowledge Units can be contracted to observe the similarity between their claims. For example, two knowledge units with contradicting claim have vanishing contractions.

5 Structured representations

Method a on benchmark 1 was successful, but not on benchmark 2. Method b was successful on benchmark 1 and 2.

This could be either represented by ϵ_1 basis vectors to each possible substitution (i.e. each combination like *Method a on benchmark 1 was successful* is associated with an atomic variable X), or in a structured representation with variables O enumerating possible methods, benchmarks etc. The knowledge would then be represented in the matrix

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} [O_{method}, O_{benchmark}] \ .$$

References

Alex Goessmann, Janina Schütte, Maximilian Fröhlich, and Martin Eigel. A tensor network formalism for neuro-symbolic AI, January 2026. URL http://arxiv.org/abs/2601.15442. arXiv:2601.15442 [cs].

Christoph Schuhmann, Gollam Rabby, Ameya Prabhu, Tawsif Ahmed, Andreas Hochlehnert, Huu Nguyen, Nick Akinci, Ludwig Schmidt, Robert Kaczmarczyk, Sören Auer, Jenia Jitsev, and Matthias Bethge. Project Alexandria: Towards Freeing Scientific Knowledge from Copyright Burdens via LLMs, April 2025. URL http://arxiv.org/abs/2502.19413. arXiv:2502.19413 [cs.LG].